Wormhole Aggregation: Recursive Zero-Knowledge Settlement for Post-Quantum Blockchains

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Abstract

Post-quantum signature schemes impose large public keys and signatures, increasing on-chain costs while still leaking transaction structure. We present *Quantus Wormhole*, a private settlement protocol where users deposit to addresses derived from secrets and later prove in zero knowledge that the deposit exists, hasn't been spent, and satisfies fee rules. Recursive aggregation amortizes verification across many withdrawals, while dummy padding and local shuffling enlarge the anonymity set for balance-linking adversaries. We formalize the withdrawal relation, analyze privacy guarantees, prove security properties, and report proving times for batch sizes up to 49.

1 Introduction

The post-quantum transition poses a dilemma for blockchains. Shor's algorithm breaks elliptic-curve signatures [16], but post-quantum replacements like ML-DSA impose keys and signatures an order of magnitude larger [8]. A naïve swap increases bandwidth and state growth while doing nothing for privacy. Every transaction still reveals sender, recipient, and amount.

One response is to verify post-quantum signatures inside a zero-knowledge circuit and aggregate many such proofs, compressing the on-chain footprint. But this requires implementing post-quantum signature verification in-circuit, which is expensive.

This paper takes a different approach: rather than proving signature validity, users prove knowledge of a secret tied to an unspent deposit. This idea appears in Ziggy [4], a STARK-based signature scheme where signers prove knowledge of a hash preimage. We apply the same principle to blockchain settlement, building on EIP-7503's "wormhole" concept [7]: funds are sent to addresses derived from secrets and later withdrawn by proving knowledge of that secret. This sidesteps post-quantum signature verification entirely. The circuit only needs hash functions and storage-proof verification. We add recursive proof aggregation for scalable settlement [14].

Contributions

1. A formal specification of the wormhole-address withdrawal relation, covering nullifier derivation, authenticated state inclusion, and fee constraints.

2. A two-layer recursive aggregation architecture where the first layer performs privacy-sensitive dummy padding and shuffling on the client, while subsequent layers can be delegated.

3. Combinatorial analysis of anonymity set size under the layer-0 grouping rules.
4. Security proofs for deposit binding, one-time withdrawal, and value conservation, with explicit separation of proof-level and chain-level assumptions.
5. Benchmarks showing sub-second proving and sub-20ms verification for batches up to 49 withdrawals.

2 Background and Design Goals

2.1 Wormholes and Privacy Pools

EIP-7503 proposes *zero-knowledge wormholes*: deposit to secret-derived addresses, then prove knowledge of the secret to withdraw [7]. We extend this concept with post-quantum proving and recursive aggregation with dummy padding for privacy.

Related systems include privacy pools and shielded ledgers like Zcash and Tornado Cash [3, 6, 7, 11]. Unlike fully shielded ledgers, Quantus proves against native chain state without a separate note commitment tree. Importantly, no ciphertexts are stored on-chain, so a future break in cryptographic assumptions cannot retroactively compromise transaction privacy.

2.2 Transparent Recursion

We use Plonky2, combining PLONK arithmetization with FRI commitments for recursive verification without trusted setup [1, 10]. Poseidon2 provides efficient in-circuit hashing over prime fields [5].

2.3 Design goals

The protocol aims to satisfy four simultaneous goals:

1. **Amortized Verification:** The marginal on-chain cost of verifying an additional withdrawal should be small.
2. **Deposit Binding:** Every withdrawal must be bound to a real deposit recorded in authenticated chain state.
3. **Replay Prevention:** A deposit should not authorize multiple withdrawals; nullifiers must uniquely bind proofs to deposits.
4. **Meaningful Privacy:** The chain should reveal only the public information needed for settlement: exits, amounts, shared fee parameters, nullifiers, and a block hash. Funding accounts, secrets, and raw storage proofs should remain private.

3 System, Adversary, and Trust Model

3.1 Roles

- **User / Prover:** Generates a secret, derives a wormhole address, deposits funds, and later produces a zero-knowledge proof to withdraw. First-layer aggregation should be performed locally to preserve privacy.

- **Aggregator:** Combines multiple proofs into a single aggregate proof. This role may be the user, a third-party service, or a block producer. Higher-layer aggregation can be delegated since it operates only on public outputs.
- **Blockchain / Verifier:** Validates the final proof, checks nullifier freshness, and settles withdrawals to the specified exit accounts.

3.2 Adversarial capabilities

We assume a probabilistic polynomial-time adversary that can:

- observe the full chain and all public proof outputs,
- submit arbitrary deposits,
- perform timing analysis, and
- attempt to replay or malform proof submissions.

The adversary cannot:

- break the collision resistance of the hash function,
- break the knowledge soundness of the proof system, or
- violate the chain's block-history availability.

3.3 Trust assumptions

The protocol relies on the following assumptions.

1. **Hash Security:** The Poseidon2 instantiation used by the system remains collision resistant and preimage resistant for the relevant security level [5].
2. **Proof Soundness:** The configured Plonky2 recursion stack remains knowledge sound under its standard assumptions [1, 10].
3. **Authenticated History:** The blockchain can check that the public block hash and block number correspond to a valid block in chain history.
4. **Witness Availability:** Honest users can obtain the storage proof and block-header data needed to build a witness for their deposit.

4 Protocol Overview

The protocol consists of five stages.

1. Secret generation and deposit. The user samples a secret s and derives a wormhole address

$$\mathsf{WA}(s) = H(H(\mathsf{salt}_{\mathsf{wh}} \parallel s)).$$

Because this address is derived from a hash rather than a public key, no one can spend from it via ordinary signature-based authorization—funds sent here are effectively burned. The user then deposits to this address through an ordinary on-chain transfer, creating a record that can later be authenticated inside the circuit.

2. Witness Acquisition: To withdraw, the user gathers the deposit record, a storage proof, and block header data. These form the private witness.

3. Leaf Proof Generation: The user chooses up to two exit outputs, computes a nullifier

$$\mathsf{Null}(s, c) = H(H(\mathsf{salt}_{\mathsf{null}} \parallel s \parallel c)),$$

and produces a leaf proof. The proof reveals the nullifier, the exits, the output amounts, the asset identifier, the fee basis points, the public block hash, and the block number, while keeping the secret, funding account, storage proof, and transfer-specific raw details private. Leaf proofs are not zero-knowledge. They are only seen by the user who generated them.

4. First-Layer Aggregation: The user batches leaf proofs that share the same `block_hash`, pads with dummy leaves if needed, shuffles the order, and produces a zero-knowledge wrapper proof that forwards grouped exits and nullifiers. This step should be performed locally to preserve privacy.

5. Settlement: The blockchain verifies the aggregate proof, checks that the public `block_hash` matches the referenced block number, verifies that the fee parameter matches the configured rate, rejects reused nullifiers, and settles the exits.

5 Formal Specification and Interfaces

5.1 Leaf public inputs

The leaf proof exposes the following public fields:

- `asset_id`: Token identifier.
- `output_amount_1`, `output_amount_2`: Exit amounts (secondary is zero if unused).
- `fee_bps`: Fee parameter in basis points.
- `nullifier`: Replay-prevention tag derived from secret and counter.
- `exit_account_1`, `exit_account_2`: Destination accounts.
- `block_hash`: Commitment to block header fields.
- `block_number`: Height of the referenced block.

5.2 Leaf private witness

The private witness includes:

- `secret`: User secret from which nullifier and wormhole address are derived.
- `counter`: Per-deposit counter used in nullifier derivation.
- `funding_account`: Source account of the deposit.
- `wormhole_account`: Derived wormhole address $\mathsf{WA}(s)$.
- `storage_proof`: Inclusion proof for the deposit record.

- `block_header`: Block header fields for commitment verification.
- `input_amount`: Deposited amount before fee deduction.

5.3 Leaf relation

Let $x = (a, v_1, v_2, f, n, \alpha_1, \alpha_2, h_B, b)$ denote the public input tuple and let w denote the witness described above. The leaf relation $\mathcal{R}_{\text{leaf}}$ holds when there exists a witness w such that:

(C1) **Nullifier Correctness:** $n = \mathsf{Null}(s, c)$.

(C2) **Wormhole-Address Correctness:** $\alpha_{\text{wh}} = \mathsf{WA}(s)$.

(C3) **Authenticated Inclusion:** The storage proof verifies that the deposit record exists in chain state.

(C4) **Header Binding:** The block-header fields hash to the public commitment h_B, and the public block number matches.

(C5) **Fee Constraint:** The disclosed exits satisfy

$$(v_1 + v_2) \cdot 10000 \leq v_{\text{in}} \cdot (10000 - f).$$

5.4 On-chain verification

The proof system does not, by itself, settle a withdrawal. The blockchain must also:

- verify the referenced block exists in chain history
- check the fee parameter matches the configured rate
- reject previously-seen nullifiers
- credit exit accounts without increasing total supply

6 Recursive Aggregation Architecture

6.1 First-layer aggregation

The first aggregation layer is privacy-sensitive and should be executed locally. A batch of up to N leaf proofs is aggregated as follows:

- All real leaves must share the same `block_hash`, `asset_id`, and `fee_bps`.
- If fewer than N real leaves are available, the batch is padded with dummy leaves.
- The leaf order is shuffled before wrapping.
- Exits to identical accounts are grouped into a single summed output.
- Real nullifiers are forwarded; dummy nullifiers are replaced by hashing prover-supplied random preimages, preventing a malicious prover from choosing a dummy nullifier that collides with a real one.

A delegated layer-0 service would observe raw child proofs before padding and shuffling, collapsing much of the anonymity set quantified in section 7.7.

6.2 Higher-layer aggregation

Higher layers aggregate already-wrapped proofs. Since they operate only on public outputs, they can be safely delegated and do not require zero-knowledge, enabling faster proving.

7 Security and Privacy Analysis

This section separates *proof-level* statements from *system-level* statements. Proof-level statements concern the relations enforced by the circuit. System-level statements additionally rely on chain-level checks: block validation, fee parameter matching, nullifier rejection, and correct settlement.

7.1 Authenticated deposit records

To tighten the security argument, it is useful to name the object that the proof is actually certifying.

Definition 1 (Authenticated deposit record)**.** *An* authenticated deposit record *is a tuple*

$$d = (a, v_{\text{in}}, c, \alpha_{\text{fund}}, \alpha_{\text{wh}}, h_B, b)$$

for which there exists witness data w such that $(x, w) \in \mathcal{R}_{\text{leaf}}$, where x exposes asset identifier a, block hash h_B, block number b, and exits satisfying the fee constraint against v_{in}.

A *real* leaf is a leaf whose public `block_hash` is nonzero. A *dummy* leaf is a padding object identified by the sentinel `block_hash` $= 0^{32}$.

7.2 Soundness and deposit binding

Theorem 1 (Leaf soundness)**.** *Assume the leaf proof system is knowledge sound. For every accepted leaf proof (x, π), there exists an extractor that outputs a witness w such that $(x, w) \in \mathcal{R}_{\text{leaf}}$.*

Proof sketch. This is the standard consequence of knowledge soundness for the leaf relation. The extractor outputs values for the secret, counter, funding account, wormhole account, storage-proof witness, header fields, and input amount that satisfy conditions (C1)–(C5) from section 5. No further claim is made here about whether the committed block actually exists in canonical chain history; that is deferred to the on-chain check of the public `block_hash`/`block_number` pair. □

Corollary 2 (Authenticated-deposit binding)**.** *Assume leaf soundness and correct on-chain validation of the public block hash. Then every accepted real leaf proof binds to an authenticated deposit record and certifies that its disclosed exits satisfy the protocol fee rule for that record.*

Proof. By leaf soundness, an accepting real leaf yields a witness satisfying $\mathcal{R}_{\text{leaf}}$. Conditions (C3) and (C4) bind the proof to an authenticated transfer record under the state root committed by the structured header fields, while condition (C5) enforces the fee inequality. On-chain validation upgrades the public header commitment from a merely self-consistent header witness to a real block in chain history. □

7.3 First-layer wrapper soundness

Theorem 3 (First-layer wrapper soundness)**.** *Assume the recursive verifier gadgets and the wrapper circuit are sound. Let Y be the public output of an accepted first-layer wrapper proof of arity N. Then there exists a list of child instances*

$$((x_1, \pi_1, \delta_1), \ldots, (x_N, \pi_N, \delta_N)),$$

where $\delta_i \in \{0, 1\}$ indicates whether child i is dummy, such that:

(W1) if $\delta_i = 0$, then π_i is an accepted leaf proof on public input x_i;

(W2) if $\delta_i = 1$, then child i is recognized by the dummy sentinel `block_hash` $= 0^{32}$*;*

(W3) all non-dummy children share the same public header tuple exposed by Y: a common block hash, asset identifier, fee parameter, and block number;

(W4) the exit slots exposed by Y are exactly the result of applying the wrapper's grouping rule to the multiset of non-dummy child exits; and

(W5) the nullifier region of Y contains every real child nullifier, while dummy positions are filled by wrapper-supplied replacement values.

Proof sketch. Soundness of the recursive verifier gadgets ensures that every accepted child proof referenced by the wrapper witness is itself valid. The wrapper then recomputes its public outputs from those children subject to explicit conditional checks: non-dummy children must agree on the batch `block_hash`, `asset_id`, and `volume_fee_bps`; exit slots are formed by the implemented grouping-and-zeroing logic; and nullifiers are forwarded child-by-child with dummy substitution in positions marked by the sentinel. The shared `block_number` claim is obtained by combining wrapper soundness with leaf soundness: each real child is itself a sound leaf proof satisfying condition (C4), so a common `block_hash` implies a common committed header and hence a common `block_number`. Hence any accepting wrapper proof can be explained by a concrete list of real and dummy children satisfying (W1)–(W5). □

Lemma 4 (Dummy-padding safety)**.** *Assume the designated dummy template exposes zero-valued exits, and the blockchain only settles the public exits produced by the wrapper. Then dummy padding cannot increase settled value. Random replacement of dummy nullifiers affects only accidental-collision probability and therefore liveness, not soundness.*

Proof. Under the stated dummy-template assumption, a dummy child contributes no positive exit amount to the grouped wrapper outputs. Replacing its nullifier with a fresh filler does not create an authenticated deposit record or any additional settled value; at worst, an accidental nullifier collision causes rejection of the aggregate. Thus dummy padding cannot authorize value that was absent from the real children. □

7.4 One-time withdrawal and double-spend resistance

Theorem 5 (One-time withdrawal per authenticated deposit record)**.** *Assume:*

(D1) the blockchain rejects reused nullifiers, including duplicates within one candidate aggregate submission;

(D2) the authenticated transfer record fixes the transfer counter c used in nullifier derivation; and

(D3) the hash function used in $\mathsf{WA}(\cdot)$ *and* $\mathsf{Null}(\cdot,\cdot)$ *is collision resistant and second-preimage resistant for the relevant domain-separated inputs.*

Then no probabilistic polynomial-time adversary can obtain two accepted real withdrawals from the same authenticated deposit record except with negligible probability.

Proof sketch. Suppose two accepted real withdrawals correspond to the same authenticated deposit record. By authenticated-deposit binding, extraction yields witnesses (s, c) and (s', c') together with the same authenticated record

$$(a, v_{\mathrm{in}}, c, \alpha_{\mathrm{fund}}, \alpha_{\mathrm{wh}}, h_B, b).$$

Because the record fixes the counter, we have $c = c'$. Because the record also fixes the wormhole address, both witnesses satisfy

$$\mathsf{WA}(s) = \alpha_{\mathrm{wh}} = \mathsf{WA}(s').$$

If $s \neq s'$, the adversary has found a collision or second preimage for the wormhole-address derivation. Therefore $s = s'$ except with negligible probability. But then

$$\mathsf{Null}(s, c) = \mathsf{Null}(s', c'),$$

so both withdrawals expose the same nullifier. The blockchain rejects the second occurrence by assumption (D1). Hence one authenticated deposit record can underwrite at most one accepted real withdrawal. □

The theorem above is intentionally phrased in terms of an *authenticated deposit record*, not an informal notion of "the same funds." If a future protocol version supports explicit partial-withdrawal state transitions, that stronger state model should be proved separately rather than inferred from the present one-shot nullifier discipline.

7.5 Operational unspendability and spend-path exclusivity

The protocol needs one assumption external to the circuit: funds sent to a wormhole address should not be directly spendable through ordinary signature-based authorization.

Proposition 6 (Spend-path exclusivity under the chain's address discipline)**.** *Let* $\mathsf{Addr} : \mathcal{PK}_\Sigma \to \{0,1\}^{256}$ *denote the chain's ordinary address derivation from externally controlled public keys. Suppose:*

(U1) every direct spend accepted by the base chain must be authorized by a valid public key $pk \in \mathcal{PK}_\Sigma$ *whose address is* $\mathsf{Addr}(pk)$*; and*

(U2) no efficient adversary can find $pk \in \mathcal{PK}_\Sigma$ *such that* $\mathsf{Addr}(pk) = \mathsf{WA}(s)$ *for a target wormhole address, except by breaking the collision or preimage resistance of the derivation.*

Then value sent to $\mathsf{WA}(s)$ *can leave the system only through the proof-authorized settlement path.*

Proof sketch. Any direct spend must use some valid external credential pk by assumption (U1). If such a spend exists from $\mathsf{WA}(s)$, then $\mathsf{Addr}(pk) = \mathsf{WA}(s)$, contradicting (U2) except with negligible probability. Therefore the only remaining spend path is the proof-authorized one. We treat spend-path exclusivity as a system assumption about the surrounding chain interface rather than as a theorem of the circuit in isolation. □

7.6 Value conservation

Lemma 7 (Per-leaf fee bound)**.** *For every accepted real leaf proof, the sum of its two public output amounts is at most the authenticated input amount net of the public fee parameter.*

Proof. This is exactly condition (C5) of $\mathcal{R}_{\text{leaf}}$, obtained by extraction from leaf soundness. □

Lemma 8 (Wrapper amount preservation)**.** *Assume the first-layer wrapper groups identical exit accounts by summing their amounts into one slot and zeroing duplicate slots. Then the sum of the forwarded exit amounts equals the sum of the exit amounts of the real child proofs.*

Proof. The wrapper's grouping step is a partition of the multiset of real child exits by destination account. Replacing each partition class by a single slot whose amount is the class sum preserves the total amount. Zeroing duplicate slots removes representation duplicates, not value. □

Theorem 9 (Global value conservation)**.** *Assume:*

(V1) leaf soundness,

(V2) first-layer wrapper soundness,

(V3) blockchain rejection of reused nullifiers, and

(V4) blockchain settlement that applies exactly the forwarded public exit amounts.

Then the total value settled by any accepted first-layer aggregate proof is at most the total value of its underlying authenticated deposit records after protocol fees. Across multiple accepted proofs, no authenticated deposit record is counted more than once.

Proof sketch. By first-layer wrapper soundness, an accepted aggregate expands to a multiset of real child leaves and dummy padding. By dummy-padding safety, dummies contribute no settled value. By authenticated-deposit binding and the per-leaf fee bound, each real child contributes at most its authenticated deposit amount net of fees. By wrapper amount preservation, the sum of forwarded exit amounts equals the sum of the real child exit amounts. Finally, one-time withdrawal per authenticated deposit record rules out counting the same authenticated record twice via multiple accepted nullifiers. Hence settled value never exceeds the aggregate authenticated input value net of fees. □

7.7 Layer-0 anonymity sets and balance-linking difficulty

The first aggregation layer does not hide the final exits or their amounts, but it does make leaf-level linkage more ambiguous. The key point is that the layer-0 wrapper is many-to-one: multiple ordered batches of real and dummy leaves can induce the same public aggregate once dummy nullifiers are replaced and duplicate exits are grouped.

Definition 2 (Layer-0 anonymity set)**.** *Fix a first-layer public output Y. The* anonymity set $\mathcal{A}(Y)$ *is the set of ordered batches of real leaves and dummy leaves, together with per-leaf two-output assignments, that are consistent with the leaf fee rule and that produce Y under the layer-0 padding, shuffle, grouping, and dummy-nullifier-substitution logic.*

A balance-linking adversary who sees only Y must distinguish among the elements of $\mathcal{A}(Y)$. By contrast, a delegated layer-0 aggregator that receives the client's raw leaf proofs before padding and shuffling learns one concrete preimage of Y and therefore collapses this ambiguity. This is why layer-0 should be performed client-locally.

Proposition 10 (Client-local layer-0 dominates delegated layer-0 for linkage privacy)**.** *Fix any first-layer public output Y. A downstream observer when layer-0 is performed client-locally receives only Y, whereas a delegated layer-0 aggregator receives both Y and one specific preimage $B \in \mathcal{A}(Y)$. Under a uniform prior over $\mathcal{A}(Y)$, the optimal exact leaf-to-output linkage success probability is $1/|\mathcal{A}(Y)|$ for the former view and 1 for the latter.*

Proof. The downstream observer receives only Y, so any exact linkage guess is a guess of which element of $\mathcal{A}(Y)$ generated the aggregate. Under the stated prior, the optimal strategy succeeds with probability $1/|\mathcal{A}(Y)|$. A delegated layer-0 aggregator, however, observes the raw child proofs before padding and shuffling, so it knows the realized preimage B directly. □

The size of $\mathcal{A}(Y)$ grows combinatorially whenever multiple hidden child outputs collapse into one grouped exit.

Proposition 11 (Combinatorial lower bound from grouped exits)**.** *Let $\Delta = 0.01$ be the circuit quantization unit. Fix a grouped exit account a in Y whose public forwarded amount is $S_a \in \Delta\mathbb{Z}_{>0}$. If that total arose from $r_a \geq 1$ positive real child outputs to the same account, then the number of positive quantized decompositions of S_a into r_a hidden child contributions is*

$$\binom{S_a/\Delta - 1}{r_a - 1}.$$

Consequently, if A^+ denotes the set of grouped exit accounts that received contributions from more than one real child output, then

$$|\mathcal{A}(Y)| \geq \prod_{a \in A^+} \binom{S_a/\Delta - 1}{r_a - 1},$$

with additional multiplicative ambiguity coming from hidden dummy placement, private batch order, and the assignment of at most two outputs to each real leaf.

Proof. For a fixed account a, write $S_a = m_a\Delta$ with $m_a \in \mathbb{Z}_{>0}$. Counting positive integer solutions to

$$x_1 + \cdots + x_{r_a} = m_a$$

gives the classical stars-and-bars count $\binom{m_a-1}{r_a-1}$. Each such solution corresponds to a distinct quantized decomposition of the grouped amount into hidden child outputs. Independent grouped accounts multiply, yielding the stated lower bound. Any uncertainty about which leaves were dummy, how the client ordered the batch before wrapping, or which of a leaf's two output positions carried a given contribution can only enlarge the anonymity set. □

7.8 Privacy boundary

The proof system hides secrets, funding accounts, and storage proofs—these are witness data, not public inputs. However, output amounts, exit accounts, nullifiers, and block references are public. Layer-0 padding enlarges the anonymity set but does not provide full unlinkability; amount patterns, address reuse, and timing correlation remain potential vectors for deanonymization.

8 Evaluation

We report wall-clock times for the current implementation.

N leaf proofs	Build (s)	Prove (s)	Verify (ms)
2	9.20	3.83	4.50
4	7.46	3.72	4.71
8	11.30	7.32	5.53
9	11.20	7.97	5.17
16	16.69	13.47	9.26
25	16.02	16.34	11.71
32	29.45	30.84	7.51
36	28.21	32.42	7.26
49	30.93	26.66	19.58

Table 1: Measured wall-clock times for the implemented first recursive aggregation layer. All runs use same-`block_hash` dummy leaf proofs and report the current refactored implementation only.

8.1 Methodology

Benchmarks were executed on an Apple M2 Max with 12 CPU cores and 32 GB of RAM, running macOS 26.3.1 with `rustc 1.90.0` and `cargo 1.90.0`. To reduce interference from prior builds, runs used scratch-local `HOME`, `CARGO_HOME`, `TMPDIR`, and `CARGO_TARGET_DIR` rooted under `/tmp`.

For the implemented first aggregation layer, we evaluated same-`block_hash` batches with $N \in \{2, 4, 8, 9, 16, 25, 32, 36, 49\}$ leaf proofs. For the next recursive layer, we fixed each layer-0 proof to aggregate eight leaves and varied the number of input layer-0 proofs $M \in \{2, 4, 8, 16, 32\}$.

All first-layer measurements use repeated identical dummy leaf proofs, so they remain within the implemented same-block / shared-`block_hash` protocol surface; they do not measure heterogeneous real proofs drawn from a live block. The second-layer runs likewise use repeated dummy-derived layer-0 proofs with full batches only.

Reported build times are builder wall-clock times. For layer 0, build includes leaf and layer-0 artifact generation plus configuration hashing. For layer 1, build measures layer-1 artifact generation only, with prerequisite layer-0 artifacts generated separately. Reported prove times include backend load/deserialization and the `commit()` stage. Reported verify times isolate verification from proof generation, but still include verifier load/deserialization from serialized circuit binaries.

8.2 First-layer aggregation results

Across the tested range, first-layer build times lie between 7.46 s and 30.93 s, proving times between 3.72 s and 32.42 s, and verification times between 4.50 ms and 19.58 ms. Verification remains below 12 ms for all tested sizes up to $N = 36$, with the largest tested batch ($N = 49$) verifying in 19.58 ms. These results indicate that the refactored aggregation path preserves a very small verifier footprint even when the proving side grows into the tens-of-seconds regime.

8.3 Second-layer aggregation results

For second-layer aggregation with eight leaves per input layer-0 proof, build times range from 5.54 s to 25.78 s, proving times from 3.52 s to 21.32 s, and verification times from 5.18 ms to 6.88 ms across $M \in \{2, 4, 8, 16, 32\}$. Verification is effectively flat across the tested fan-ins, while proving and artifact generation scale with the number of recursive inputs.

M layer-0 proofs	Leaves per layer-0 proof	Build (s)	Prove (s)	Verify (ms)
2	8	5.56	3.52	5.18
4	8	5.54	4.23	5.34
8	8	9.22	7.21	5.94
16	8	13.27	13.86	6.39
32	8	25.78	21.32	6.88

Table 2: Measured wall-clock times for the current second-layer aggregation interface. These measurements are head-only because the explicit public layer-1 surface is introduced by the refactor.

8.4 Forwarded payload and comparison basis

The wrapper-level public-output accounting is unchanged by the refactor. For a first-layer batch of size N, the forwarded semantic payload remains $14 + 8/N$ field elements per settled leaf, excluding zero padding in the wrapper's fixed-width public-input layout. This remains the correct accounting basis for any later throughput comparison against direct post-quantum signatures.

The current timing data establish the wall-clock behavior of the refactored aggregation flow.

9 Limitations

Same-block scoping. Layer-0 batches proofs sharing the same `block_hash`. Cross-block aggregation is future work.

Two-output limit. Each leaf supports two exits (spend + change), not general multi-recipient batching.

Amount quantization. Amounts are quantized to 0.01-unit resolution to reduce circuit cost.

Privacy leakage. Output amounts, exit addresses, and timing are public. Layer-0 padding enlarges the anonymity set (section 7.7), but this is plausible deniability, not full shielding.

10 Related Work

Wormholes and privacy pools. EIP-7503 proposes proof-of-burn reminting via secret-derived addresses [7]. Privacy Pools adds compliance-aware privacy sets [3]. Quantus implements recursive proofs over authenticated chain state.

Shielded systems. Zcash provides strong privacy through shielded notes; Tornado Cash popularized mixer-style privacy on Ethereum [6, 11]. Quantus offers lighter-weight private settlement anchored in native chain state.

Recursive proofs. Halo and Plonky2 are standard recursive proof systems [2, 10]. Plonky3 provides Poseidon2 primitives with a separate recursion track [12, 13]. We use Plonky2 with Poseidon2 for recursive composition.

Hash-based authorization. The Ziggy signature scheme [4] uses a ZK-STARK to prove knowledge of a hash preimage, replacing lattice-based signatures with hash-based proofs. Our wormhole construction applies a similar idea to blockchain settlement: users prove knowledge of a secret whose hash determines their deposit address.

Post-quantum blockchains. ML-DSA adoption raises questions about transaction size and verification cost [8]. Quantus compresses authorization into ZK proofs rather than publishing large signatures on-chain.

11 Conclusion

Wormhole addresses let a post-quantum blockchain move authorization off-chain while preserving authenticated settlement. Users deposit to secret-derived addresses, prove the deposit exists and hasn't been spent, and reveal only settlement outputs. Recursive aggregation amortizes verification. We formalized the leaf relation, analyzed layer-0 privacy, proved security properties, and reported benchmark timings.

A Privacy-score heuristic

A wallet-side heuristic (not a formal guarantee): for candidate deposits $\mathcal{D}$ and withdrawal amount V,

$$\mathsf{Score}(V, \mathcal{D}) = \log_2 \left| \left\{ S \subseteq \mathcal{D} : \sum_{d \in S} d \cdot (1 - f) \approx V \right\} \right|.$$

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